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Practical Machine Learning

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Project 2: Classification

a. Introduction

Electricity markets are unusually attractive settings for short-term prediction because supply and demand must be balanced continuously, storage is limited, and small shifts in system conditions can affect prices and trading opportunities very quickly. In the Great Britain (GB) balancing system, periods can be thought of as “short” when the system needs additional energy and “long” when the system has excess energy. That makes the direction of system imbalance economically meaningful for anyone interested in directional trading decisions.

Short-term electricity trading refers to the buying and selling of power for delivery in the near future, often within the same day or even the same settlement period, in order to respond to constantly changing conditions on the grid. It happens because electricity demand and supply must remain balanced in real time, yet both can shift quickly as consumption changes, generators go offline, or renewable output rises and falls. These short term trading markets allow traders, generators, suppliers, and other participants to adjust positions close to delivery, helping correct imbalances before they become more costly for the system operator to manage directly. In that sense, short-term trading helps solve a core coordination problem in electricity systems. The problem is that power cannot be economically stored at large scale in the same way as many other commodities, so deviations between expected and actual conditions must be absorbed and adjusted to continuously. The result, when these markets function well, is a grid that is easier and cheaper to balance, less exposed to extreme pricing, and better able to integrate variable generation while maintaining reliability. In practice, this means traders and energy providers are continually buying, selling, and rebalancing power positions close to delivery as updated information on demand, generation, outages, and prices arrives. Because positions must be set before final prices and system conditions are fully known, accurate prediction is essential for traders to anticipate whether the market is likely to tighten or loosen and to enter positions that will be profitable rather than exposed.

This project asks whether machine learning models can predict whether the GB electricity system will be strongly short or strongly long one hour ahead. The positive class in this project is a strongly short period, coded as one, because correctly identifying tightening system conditions is the more decision-relevant outcome for short-term trading and market positioning. A useful classifier could therefore support a simple directional trading rule in which periods predicted to

be strongly short are interpreted as buy-oriented signals, while periods predicted to be strongly long are interpreted as sell-oriented signals. The core empirical question is whether recent imbalance conditions and recent price behavior contain enough information to classify one-hour-ahead market states out of sample. To answer that question, I compare logistic regression, random forest, and neural networks. I first construct a richer forecasting dataset using lagged, rolling, and change-based measures of imbalance and prices, then evaluate the models using a chronological train/test split and shared-fold cross-validation on the training sample. The results show that predictive performance depends more on careful problem definition and informative market-state variables than on model complexity alone, although the random forest ultimately emerges as the strongest model under cross-validated selection.

b. Data Description

The dataset was constructed from publicly available Elexon settlement files. Two monthly file families were used. System Period Data (SSD) provided system-level variables including Net Imbalance Volume, System Buy Price, System Sell Price, Loss of Load Probability, and De-rated Margin. Market Index Data Period Information (MD1) provided market index prices and volumes. I downloaded monthly files covering January 2024 through April 2026 and then filtered the combined panel to actual settlement dates in that period.

A key data-cleaning issue was that the month attached to each downloaded file reflects the month in which the data was produced under the settlement process, not necessarily the month of the market observation itself. For that reason, all files were stacked first and only afterward filtered by Settlement_Date. I then kept the latest available record for each Settlement_Date and Settlement_Period combination. For market index data, I kept the APXMIDP provider because it carried the usable price series that I required.

After merging SSD and MD1 by settlement date and settlement period, the initial panel contained 40,460 half-hour observations. The baseline target variable was defined as an indicator equal to one when one-hour-ahead Net Imbalance Volume was positive and zero when it was negative. This creates a direct short-versus-long classification problem, where positive future imbalance corresponds to a short system and negative future imbalance corresponds to a long system.

The final feature set was built to reflect information available before the target period. Baseline predictors included calendar variables, one and two-period lags of Net Imbalance Volume, one-period lags of system buy and sell prices, one-period lags of market index price and volume, and selected system stress measures such as Loss of Load Probability and De-rated Margin.

The final improved sample remained reasonably balanced. In that specification, approximately 44.5 percent of observations were strongly short periods and 55.5 percent were strongly long periods. The feature set was designed to reflect information plausibly available

before the target period. To construct the improved target, I excluded observations whose absolute one-hour-ahead Net Imbalance Volume fell below the 25th percentile of the absolute imbalance distribution, thereby focusing the classification problem on stronger short and long states. In addition to calendar variables and basic lags, I engineered two main groups of new features from the raw variables: first, additional lagged, change-based, and rolling measures of Net Imbalance Volume and market prices; and second, cyclical time-of-day variables that help the model capture repeating daily patterns. Summary statistics for the core market variables are reported in *Table 1*.

Figure 1

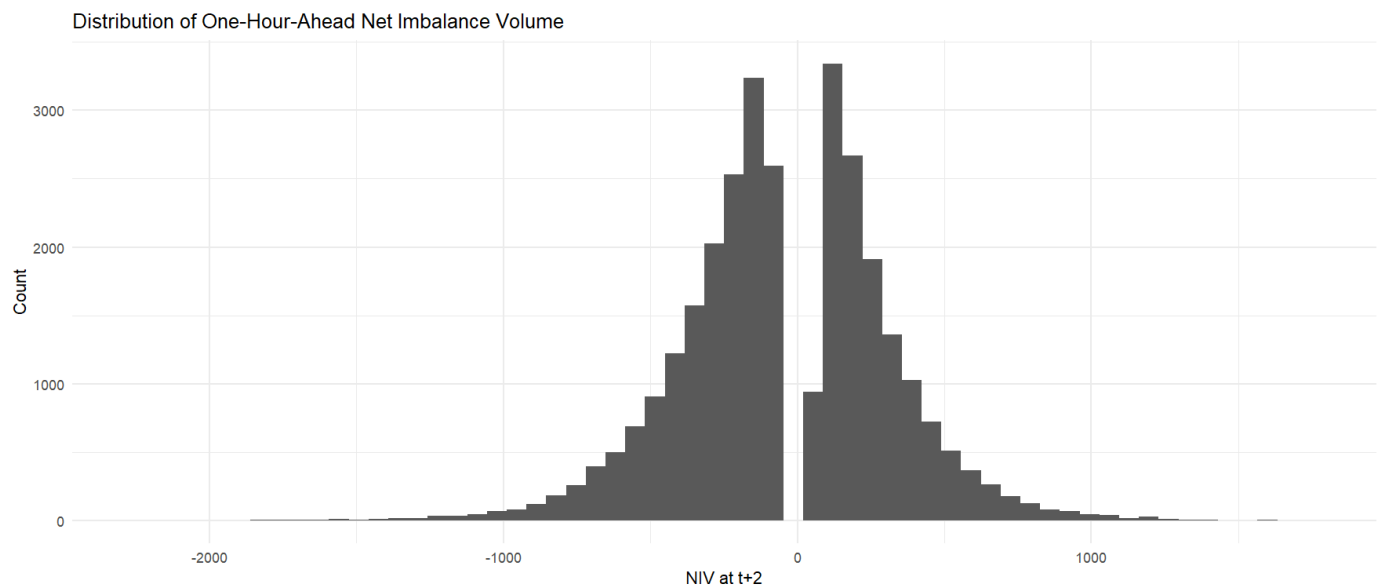


Figure 1. Distribution of one-hour-ahead Net Imbalance Volume (niv_tplus2). This figure shows a histogram of the one-hour-ahead imbalance variable and illustrates the concentration of observations near zero. *Figure 1* serves to motivate the improved target definition, which drops ambiguous near-zero periods and focuses the classification problem on stronger short/long states.

Table 1

Table: Summary Statistics for Core Market Variables

Variable	Mean	SD	Min	Max
Net_Imbalance_Volume	-41.119	340.108	-2232.378	1733.556
System_Buy_Price	75.647	59.866	-97.920	2900.000
System_Sell_Price	75.647	59.866	-97.920	2900.000
Market_Index_Price	75.648	37.734	-102.920	1352.900
Market_Index_Volume	2389.529	859.407	0.000	8004.700

Table 1. Summary statistics for core market and system variables.

This table reports descriptive statistics for the main variables used in the analysis, including Net Imbalance Volume, System Buy Price, System Sell Price, Market Index Price, and Market Index Volume. The values show variation across settlement periods, consistent with a market characterized by rapidly changing short-run conditions.

c. Methods

Because this is a forecasting problem, model evaluation used a chronological split rather than a random split. Observations were sorted by `Settlement_Date` and `Settlement_Period`, and the first 80 percent of the sample was used for training while the final 20 percent was reserved for testing. This setup better reflects a real forecasting environment and prevents future observations from entering the training sample.

Three classification methods were compared: logistic regression, random forest, and a neural network. These methods were chosen to span three distinct model families used in the course. Logistic regression serves as an interpretable linear benchmark. Random forest provides a flexible nonlinear tree-based ensemble that can capture interactions and threshold effects without requiring a prespecified functional form. The neural network offers another nonlinear alternative, allowing the model to learn complex relationships from scaled numeric inputs.

Model selection was carried out using five-fold cross-validation on the training sample only. To ensure a fair comparison across methods, I created one shared set of folds and passed the same training partitions to each classifier. Shared folds matter because they ensure that differences in performance are driven by the models themselves rather than by different random splits of the data. AUC was used as the primary model-selection metric because it is threshold-free and therefore provides a cleaner comparison across classifiers that return probabilities.

The improved modeling specification used the refined strong short versus strong long target and a richer set of forecasting-safe predictors. These included calendar variables, lagged imbalance measures, lagged prices, first differences, rolling means, rolling standard deviations, and cyclical time features. For the random forest, hyperparameter tuning focused on the number of variables tried at each split. For the neural network, tuning focused on the hidden-layer size and weight decay. Logistic regression was estimated as a benchmark without a tuning grid. After selecting the winning model by mean cross-validated AUC, I evaluated that model alone on the held-out test set using a confusion matrix, accuracy, sensitivity, specificity, balanced accuracy, and test-set AUC.

d. Results

Table 2

Table: Cross-Validated Model Comparison

Model	Mean_CV_AUC	Mean_CV_Sensitivity	Mean_CV_Specificity
Random Forest	0.7665	0.7451	0.6384
Neural Network	0.7494	0.7361	0.6181
Logistic Regression	0.7415	0.7489	0.5940

Table 2. Cross-validated performance comparison across classification models. This table reports mean cross-validated AUC, sensitivity, and specificity for the three candidate classifiers using shared folds on the training sample. The random forest achieved the highest mean CV AUC and was therefore selected as the winning model for final evaluation on the held-out test set.

Figure 2

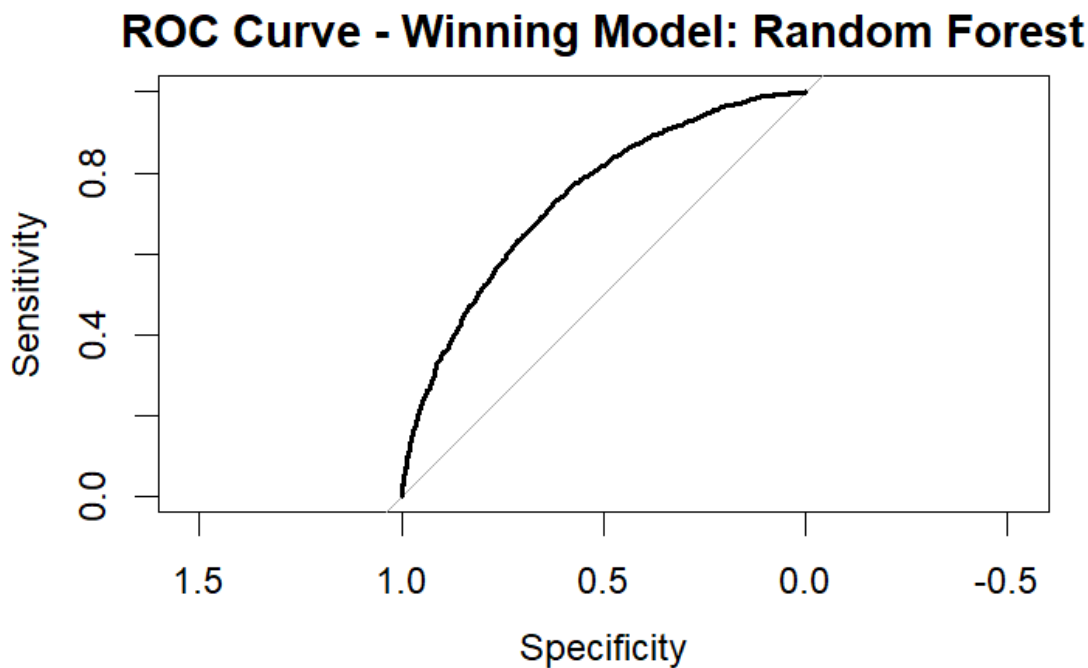


Figure 2. ROC curve for the winning random forest model on the held-out test set. The ROC curve summarizes the tradeoff between sensitivity and specificity across classification thresholds. The curve lies well above the no-skill benchmark, and the corresponding test-set AUC of 0.736 indicates that the model ranks future strong short versus strong long periods substantially better than chance.

Table 3

Table: Winning Model Test-Set Performance

Model	Test_Accuracy	Test_Sensitivity	Test_Specificity	Test_Balanced_Accuracy	Test_AUC
Random Forest	0.6756	0.6146	0.7245	0.6696	0.7362

Table 3. Held-out test-set performance of the winning random forest model.

This table reports out-of-sample classification performance for the random forest selected by cross-validated AUC. The model achieved 67.6 percent accuracy and a test-set AUC of 0.736, indicating meaningful predictive power in one-hour-ahead classification of strong short versus strong long system conditions.

Figure 3

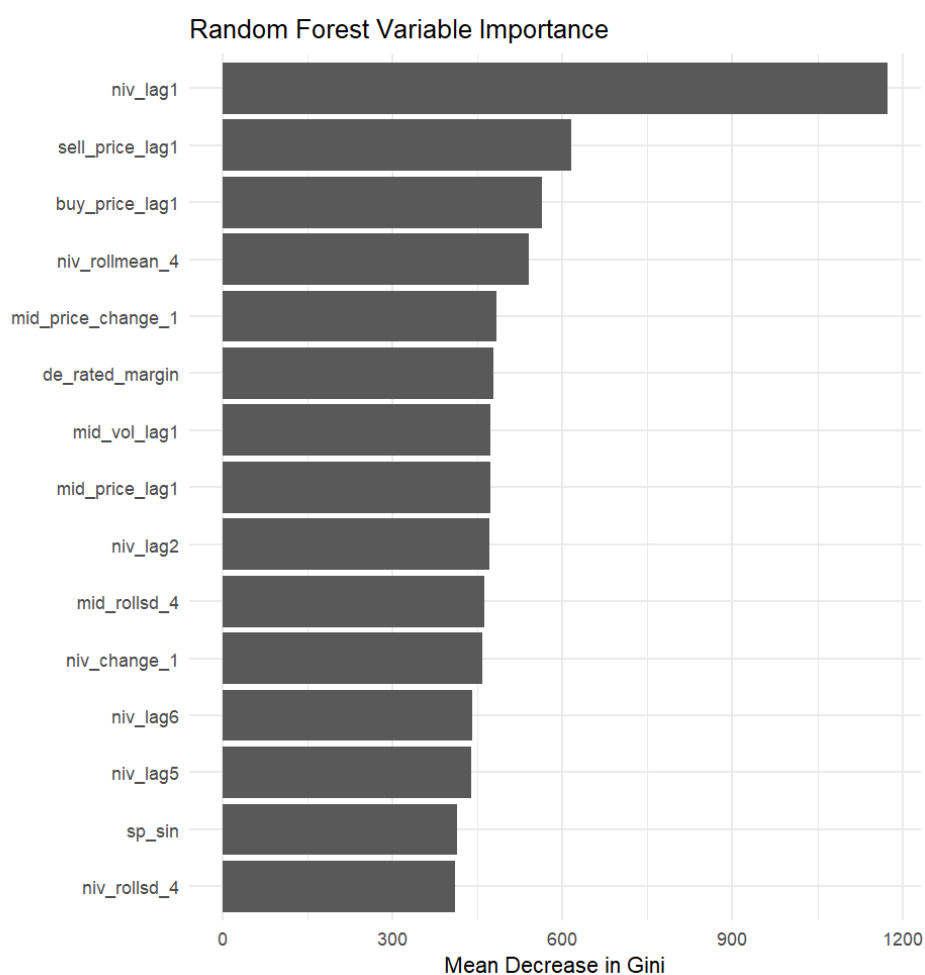


Figure 3. Variable importance from the winning random forest model.

Recent imbalance and price conditions dominate the importance rankings, with the most influential predictors including the most recent lag of Net Imbalance Volume, lagged buy and sell prices, recent average imbalance, and recent changes in market index price. This pattern

suggests that short-run market persistence and recent price dynamics are central to predicting one-hour-ahead system direction.

The baseline models performed better than the no-information benchmark, indicating that recent imbalance and price conditions do contain predictive signal. However, baseline performance was only moderate when all periods were included. Baseline accuracy clustered around 62 percent, and baseline AUC values ranged from approximately 0.666 to 0.672. These initial results suggested that the prediction problem was feasible, but also that the raw short-versus-long outcome included substantial noise.

The main improvement came from redefining the target and enriching the feature set. Once near-zero imbalance periods were removed and richer lagged, rolling, and change-based variables were added, model performance improved materially. Cross-validated model comparison on the training sample showed that the random forest performed best overall, with a mean CV AUC of 0.7665, compared with 0.7494 for the neural network and 0.7415 for logistic regression. *Table 2* presents the cross-validated comparison results, including mean CV AUC, sensitivity, and specificity for each model. Because AUC was the primary model-selection metric, the random forest was selected as the winning model.

I then evaluated the winning random forest on the held-out test set. The model achieved an accuracy of 0.6756, a sensitivity of 0.6146, a specificity of 0.7245, a balanced accuracy of 0.6696, and a test-set AUC of 0.7362. The ROC curve for the winning model is shown in Figure 2. These results indicate that the model captures meaningful short-run structure in recent imbalance and price dynamics and can classify one-hour-ahead system direction substantially better than a naive benchmark.

The variable-importance rankings from the random forest reinforce this interpretation. The most important predictors were recent imbalance and recent price conditions, especially the most recent lag of Net Imbalance Volume, lagged buy and sell prices, rolling mean imbalance, recent market price changes, and additional imbalance lags. This pattern suggests that one-hour-ahead market direction depends heavily on short-run persistence and momentum rather than on slower-moving calendar effects alone.

In deployment terms, a false positive occurs when the model predicts a strongly short period even though the system is not actually strongly short. For a trader or energy market participant, that could lead to an unnecessarily aggressive buy-oriented position or to overstating the tightness of current conditions. A false negative occurs when the model fails to identify a genuinely strongly short period. This may be more costly, since missing a tightening system could leave a trader under-positioned ahead of worsening conditions. For that reason, sensitivity to the positive class is especially important in this application, although it must be balanced against the cost of reacting to false alarms.

e. Conclusion

This project finds that one-hour-ahead strong short-versus-strong long conditions in the GB electricity market are meaningfully predictable using public settlement and market-index data. All three classification methods performed well enough to demonstrate predictive signal, but the strongest overall model under cross-validated selection was the random forest. On the held-out test set, the winning model achieved 67.6 percent accuracy and a test-set AUC of 0.7362.

The substantive lesson is that predictive performance depended more on careful target definition and variable construction than on model complexity alone. The largest gains came from redefining the outcome to focus on economically meaningful short and long states and from engineering variables that captured short-run imbalance persistence, recent price dynamics, and cyclical timing. Once the problem was specified more carefully, the nonlinear models showed a modest advantage, with the random forest emerging as the strongest classifier.

The main limitation of the analysis is that it stops at classification performance rather than full trading backtesting. A more complete project could incorporate execution rules, transaction costs, threshold tuning, and profit-based evaluation. Future work could also expand the feature set using additional demand, generation, and balancing datasets from Elexon. Even so, the current results support the idea that public electricity-market data can be used to construct useful short-horizon directional forecasts with clear relevance for market monitoring and trading applications.

f. References

Elexon. System Period Data (SSD) and Market Index Data Period Information (MD1), monthly settlement files, January 2024 to April 2026.

Pozzetti, L., & Cartlidge, J. P. (2020). Trading Electricity Markets Using Neural Networks. Proceedings of the European Modeling & Simulation Symposium.